Lesson 4 Teacher's Key

GameMaker

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

A single Collision event on obj_bullet, targeting obj_enemy:

```
// obj_bullet's Collision event with obj_enemy:
instance_destroy(other);
instance_destroy(self);
```

Level 2 - Complete Worked Solution

GameMaker's Collision event already only fires once per pair of instances actually overlapping that step, and instance_destroy() is safe to call on an instance that's already been destroyed earlier in the same step (it's simply ignored) - so even the rare case of two bullets hitting the same enemy on the same step doesn't error. No extra code is needed for Level 2.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on hit

Add sfx_hit.wav as a sound resource (keep its default name, sfx_hit), then call audio_play_sound() at the top of the Collision event, before either instance_destroy() call:

```
audio_play_sound(sfx_hit, 1, false);
instance_destroy(other);
instance_destroy(self);
```

Visual: give the enemy its own look

Set image_blend in obj_enemy's own Create event, without touching obj_bullet's code at all:

```
// obj_enemy's Create event:
image_blend = make_color_rgb(200, 25, 55);
```

Structure: keep enemy-appearance logic out of obj_bullet

obj_bullet's Collision event should only ever detect a hit and clean up (destroy the enemy, destroy itself, play the sound) - anything about how the enemy looks on its own belongs in obj_enemy's own Create event, not bolted onto obj_bullet.

Complete Level 3 solution (all 3 goals combined)

Everything above combined, across obj_bullet and obj_enemy:

```
// obj_enemy's Create event:
image_blend = make_color_rgb(200, 25, 55);

// obj_bullet's Collision event with obj_enemy:
audio_play_sound(sfx_hit, 1, false);
instance_destroy(other);
instance_destroy(self);
```

Alternate Path Notes: GML Visual

The Collision event itself is set up identically in both paths - only what's inside it differs (blocks vs. code).

The Level 3 goals map onto blocks the same way they map onto code, favoring the most specific named block available: "Play Audio" (Audio category) for sound, and "Set Instance Variable" -> Image Blend for the visual idea - Image Blend is a recognized built-in on Set Instance Variable's own dropdown, so it's the better choice over the generic Assign Variable/Set Variable. All of this stays inside obj_enemy's own events, kept separate from obj_bullet's events the same way the code path keeps the two objects separate.

- Keeping the same Structure principle (an object's own appearance/response lives with that object) whether a student is in Code or Visual reinforces that it's a design idea, not a syntax rule.